

§8.1. ~~Confidence intervals~~ Hypothesis testing: one mean (class 2)

- ① one mean: known σ
- ② one mean: unknown σ
- ③ ~~one mean: unknown σ~~
- ④ Example: decide on hypothesis using a) critical region b) p-value and c) confidence interval

① null hypothesis
 $H_0: \mu = \mu_0$
model $N(\mu, \sigma^2)$, σ^2 is known
test statistic: $\bar{x}$

Alternative hypotheses:
 options

- a) $H_1: \mu < \mu_0$ (mean decreases)
- b) $H_1: \mu > \mu_0$ (mean increases)
- c) $H_1: \mu \neq \mu_0$ (mean changes)

In all 3 cases we want to find

- I Critical regions = values $x_1, \dots, x_n$ (values for $\bar{x}$) so that H_0 is rejected
- II P-values = smallest significance level (acceptable error) on which H_0 will be ~~accepted~~ rejected

(p-value) < (significance level α) $\Rightarrow$ reject H_0

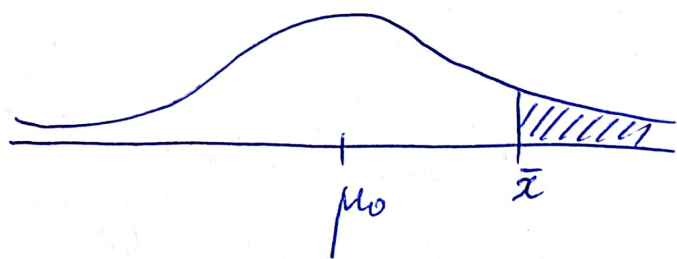
(p-value) $\geq$ (significance level α) $\Rightarrow$ do not reject (accept) H_0

- Small p-values $\rightarrow$ we tend to reject H_0
- [the smaller p-value, the less we believe in H_0 $\approx$ measure of truth of H_0]
- Tail probability; probability that statistic is at least as extreme as observed (under H_0)

(*)

(a) $H_1: \mu > \mu_0$

Shape of the critical region



Idea: reject H_0 , if $\bar{x}$ is too large

Decision rule (α):

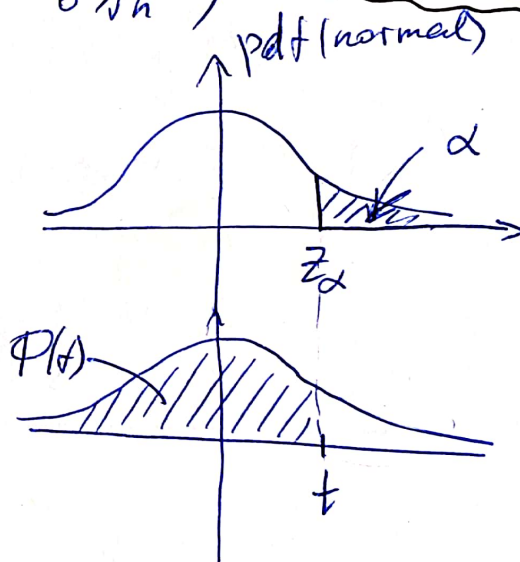
if $P(\bar{X} \geq \bar{x}) \leq \alpha$, then reject H_0

Compute it:

$$P(\bar{X} \geq \bar{x}) = P\left(\frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \geq \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}\right) = 1 - \underbrace{\Phi\left(\frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}\right)}_{\text{p-value}} \leq \alpha$$

To find such $\bar{x}$, recall:

So, $1 - \Phi(t) \leq \alpha$ iff $|t| \geq z_\alpha$

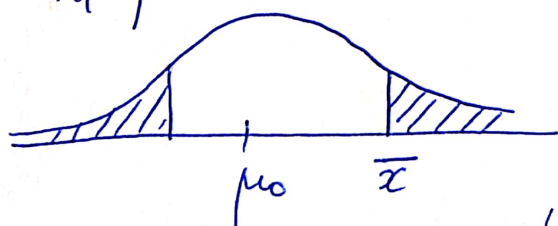


$$\frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \geq z_\alpha$$

$\bar{x} - \mu_0 \geq z_\alpha \cdot \frac{\sigma}{\sqrt{n}} \rightarrow$ all such $\bar{x}$ are in the critical region

(b) similar to (a) (exercise)

(c) $H_1: \mu \neq \mu_0$



Idea: reject H_0 , if $\bar{x}$ is too large or too small

if $P(|\bar{X} - \mu_0| > |\bar{x} - \mu_0|) \leq \alpha$, reject H_0

-3-

$$\begin{aligned}
 P(|\bar{X} - \mu_0| > |\bar{x} - \mu_0|) &= 2 P(\bar{X} - \mu_0 > |\bar{x} - \mu_0|) = \\
 &= 2 P\left(\frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \geq \frac{|\bar{x} - \mu_0|}{\sigma/\sqrt{n}}\right) = 2 \underbrace{\left(1 - \Phi\left(\frac{|\bar{x} - \mu_0|}{\sigma/\sqrt{n}}\right)\right)}_{\text{p-value}} < \alpha \\
 \underbrace{1 - \Phi\left(\frac{|\bar{x} - \mu_0|}{\sigma/\sqrt{n}}\right)} &< \alpha/2
 \end{aligned}$$

So, $|\bar{x} - \mu_0| > \underline{z_{\alpha/2}} \cdot \frac{\sigma}{\sqrt{n}} \rightarrow$ all such $\bar{x}$ are in the critical region

H_0	H_1	Critical region	p-value
a) $\mu = \mu_0$	$\mu \neq \mu_0$	$ \bar{x} - \mu_0 \geq \underline{z_{\alpha/2}} \cdot \frac{\sigma}{\sqrt{n}}$	$P(Z \geq \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}})$ $= 2(1 - \Phi(\frac{ \bar{x} - \mu_0 }{\sigma/\sqrt{n}}))$
b) $\mu = \mu_0$	$\mu > \mu_0$	$\bar{x} - \mu_0 \geq \underline{z_{\alpha}} \cdot \frac{\sigma}{\sqrt{n}}$	$P(Z \geq \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}})$
c) $\mu = \mu_0$	$\mu < \mu_0$	$\mu_0 - \bar{x} \geq \underline{z_{\alpha}} \cdot \frac{\sigma}{\sqrt{n}}$	$P(Z \leq \frac{\mu_0 - \bar{x}}{\sigma/\sqrt{n}})$

② Unknown σ^2
Same idea

a) $H_1: \mu > \mu_0$

$$\begin{aligned}
 P(\bar{X} \geq \bar{x}) &= P\left(\frac{\bar{X} - \mu}{S^2/\sqrt{n}} \geq \frac{\bar{x} - \mu}{S/\sqrt{n}}\right) = \\
 &= 1 - \underbrace{T_{n-1}\left(\frac{\bar{x} - \mu_0}{S/\sqrt{n}}\right)}_{\text{p-value}} \leq \alpha
 \end{aligned}$$

" $P(T_{n-1} \leq \frac{\bar{x} - \mu}{S/\sqrt{n}})$ for $T \sim t_{(n-1)}$

if $\frac{\bar{x} - \mu_0}{S/\sqrt{n}} \geq t_{\alpha}(n-1)$, then we reject H_0

Unknown σ	Critical region	p-value -4-
① $H_1: \mu \neq \mu_0$	$ \bar{x} - \mu_0 \geq t_{\alpha/2}(n-1) \frac{s}{\sqrt{n}}$	$2P(T \geq \frac{\bar{x} - \mu_0}{s/\sqrt{n}})$
② $H_1: \mu > \mu_0$	$\bar{x} \geq \mu_0 + t_{\alpha}(n-1) \frac{s}{\sqrt{n}}$	$P(T \geq \frac{\bar{x} - \mu_0}{s/\sqrt{n}})$
③ $H_1: \mu < \mu_0$	$\bar{x} \leq \mu_0 - t_{\alpha}(n-1) \frac{s}{\sqrt{n}}$	$P(T \leq \frac{\mu_0 - \bar{x}}{s/\sqrt{n}})$
<u>P-value alternative descriptions</u> $\otimes$		tail-probability

Example X - diameter of tumor in a mouse

$$X \sim N(\mu, 6^2)$$

$$H_0: \mu_0 = 4 \text{ mm}$$

$$H_1: \mu \neq 4$$

$n=9$ observations, significance level $\alpha=0.1$, $\bar{x}=4.3$, $s=1.2$
Accept or reject H_0 ?

$$\text{if } \frac{|\bar{x} - 4|}{s/\sqrt{9}} \geq t_{0.05}(8) = 1.86 \Rightarrow \text{reject}$$

$$\frac{0.3}{1.2/\sqrt{9}} = 0.75 < 1.86, \text{ so we accept } H_0$$

at 10% significance level

Critical region

$$\bar{x} \geq 1.86 \cdot \frac{s}{\sqrt{9}} + 4$$

$$\bar{x} \leq 1.86 \cdot \frac{s}{\sqrt{9}} - 4$$

$$[-\infty, -3.26] \text{ and } [4.744, +\infty)$$

Way 2

p-value

-5-

$$P(|T| > 0.75) \approx 2, P(T \geq 0.75) \approx$$

$$\approx 2 \cdot P(T \geq 0.706) = \underline{\underline{0.5}}$$

p-value $> \alpha \Rightarrow$ do not reject H_0 (belief is strong)

Way 3

based on confidence interval

$$\mu \in \bar{x} \pm t_{n-1, \alpha/2} \frac{s}{\sqrt{n}} \quad \text{understood}$$

$$\mu \in [3.56, 5.04]$$

$$\mu_0 = 4 \quad \checkmark$$

so we are 90% confident that μ_0 can be good $\Rightarrow$ H_0 is accepted